

Exercise 2 : Treaps

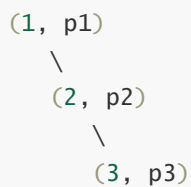
Task 1 : Possible outputs of Treap generation

There are **5** different shapes of trees if the Treap is generated by inserting the sequence of elements [1, 2, 3] in that order.

Denote the priorities of 1, 2, 3 as p_1, p_2, p_3 . They are randomly chosen when inserting the corresponding key.

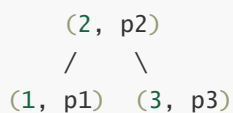
Here are the possible shapes of trees :

1. Totally right-skewed



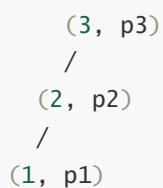
with priorities order : $p_1 < p_2 < p_3$

2. Balanced



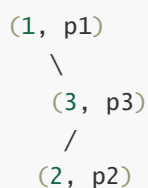
with priorities order : $p_2 < p_1, p_3$

3. Totally left-skewed



with priorities order : $p_3 < p_2 < p_1$

4. Right-heavy but 3 promoted



with priorities order : $p_1 < p_3 < p_2$

5. Left-heavy but 1 promoted

```
(3, p3)
 /
(1, p1)
 \
(2, p2)
```

with priorities order : $p3 < p1 < p2$

Task 2 : Monte Carlo approach for expected height of Treap

- [code in the wet part : `MonteCarloForTreap`]

Here's the output we get :

size	PracAvgHeight	$\log_2(\text{size})$
10	5.5900	3.3219
15	6.8000	3.9069
30	9.0400	4.9069
100	13.2100	6.6439
200	15.9600	7.6439
500	19.2800	8.9658
1000	22.3300	9.9658
2000	24.5400	10.9658
10000	31.7100	13.2877

We can see that our experimental results **does NOT** align with the theoretical expectation that the height of a Treap with n nodes is $\log n$. The practical average height is about $2 \times \log_2 n$.